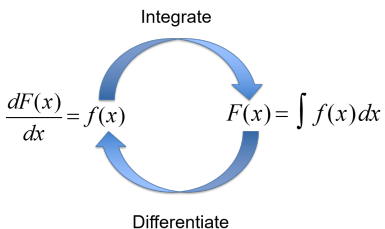


EART11200 - The Natural Scientist's Toolkit
Mathematics and Physics

Revision Lecture - Integration

- Integration is the **reverse** of differentiation.
- Since the derivative function provides us with a gradient, integration therefore provides us with a function that can be used to calculate absolute values.
- The $\int$ symbol was introduced by Gottfried Wilhelm Leibniz in the 17th century to represent summation of infinitely many infinitesimally small quantities.

e.g. An equation that calculates the rate of change of temperature in the past, present or future



e.g. An equation that calculates the temperature in the past, present or future

- We can reverse some common derivatives, we arrive at the following functions.
- The more you practice these procedures, the easier it will become to apply them without looking them up.
- Use your math's survival pack when needed!

Derivative $\frac{dy}{dx}$	**Integral** y
0	d
constant	$constant \times x + d$
x	$\frac{x^2}{2} + d$
x^n	$\frac{x^{n+1}}{n+1} + d$
e^x	$e^x + d$
ke^{kx}	$e^{kx} + d$
$\frac{1}{x}$	$ln(x) + d$
$cos(x)$	$sin(x) + d$
$sin(x)$	$-cos(x) + d$

- **Indefinite integration** (i.e. without limits)

$$\int 3x^2\,dx = x^3 + d$$

- **Definite integration** (i.e. with limits)

$$\int_0^1 3x^2\,dx = \left[x^3\right]_0^1 = 1^3 - 0^3 = 1$$

Derivative $\frac{dy}{dx}$	**Integral** y
0	d
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x	$\frac{x^2}{2} + d$
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- The equation describing radioactive decay is:

$$\frac{dN}{dt} = -\lambda N$$

where N is the number of unstable nuclei, t is time and λ is called the decay rate.

- This is a differential equation, and we used **separation of variables** to show that it has solution:

$$\int \frac{1}{N} dN = \int dt N(t) \quad = \quad N_0 \exp(-\lambda t)$$

where N_0 is the total number of nuclei in the sample at $t = 0$.

- In this course, we focussed on Newton's Second Law, which is given by the following equation:

$$F = ma$$

  where F is the net force acting on a body that has a mass m and therefore acceleration a.
- This states acceleration of an object depends on the mass of the object and the amount of force applied.
- Newton's second law can therefore be expressed as an ordinary differential equation:

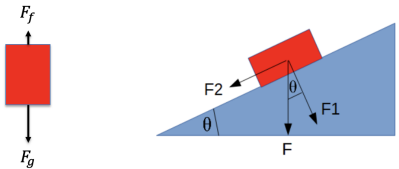
$$F = m\frac{dv}{dt} \qquad F = m\frac{d}{dt}\left(\frac{dx}{dt}\right) \qquad F = m\frac{d^2x}{dt^2} \qquad F = mv\frac{dv}{dx}$$

- By examining the forces acting on objects we can formulate ODEs:

$$F = ma \quad \Longrightarrow \quad m\frac{dv}{dt} = mg$$

$$F = ma \quad \Longrightarrow \quad m\frac{dv}{dt} = mg\sin(\theta)$$

- Using separation of variables, we integrate these ODEs to calculate behaviours of our systems.
- Frictional forces will depend on the object, the system, its mass and how quickly it is moving.



- In some cases, we cant directly integrate by using the common laws, as we are faced with more complicated combinations of functions such as **functions of functions**.
- For example, what happens if we want to integrate

$$y = \int \frac{dy}{dx} = \int (3x + 2)^4 dx$$

- This is where **integration by substitution** comes in.
- Integration by substitution is essentially the reverse process of the chain rule in differentiation.

Derivative $\frac{dy}{dx}$	**Integral** y
0	d
constant	$constant \times x + d$
x	$\frac{x^2}{2} + d$
x^n	$\frac{x^{n+1}}{n+1} + d$
e^x	$e^x + d$
ke^{kx}	$e^{kx} + d$
$\frac{1}{x}$	$ln(x) + d$
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1. Write out the integral in its simplest form:

$$y = \int_{-1}^{1} (3x + 2)^4 dx$$

2. Find a new variable (u) as a function of the old variable (x) as a substitution:

$$u = 3x + 2$$

3. Rearrange to get an expression for the old variable (x) in terms of the new variable (u):

$$x = \frac{1}{3}(u - 2)$$

4. Differentiate with respect to the new variable u:

$$\frac{dx}{du} = \frac{1}{3}$$

5. Get an expression for dx:

$$dx = \frac{du}{3}$$

6. (If needed) Find the new limits of the integral:

$$\text{When } x = -1, \quad u = 3 \times (-1) + 2 = -1$$
$$\text{When } x = 1, \quad u = 3 \times 1 + 2 = 5$$

7. Make the substitution in the original integral:

$$\int_{-1}^{1} (3x+2)^4 dx = \int_{-1}^{5} u^4 \frac{du}{3}$$

8. Integrate as usual:

$$\frac{1}{3}\left[\frac{1}{5}u^5\right]_{-1}^{5} = \frac{1}{3} \times \frac{1}{5} \times (5^5 - 1^5) = \frac{3124}{15}$$

- Hydrostatic Equation, with constant temperature requires integration via separation of variables:

$$\frac{dP}{dz} = -\frac{g}{R_\alpha T}P \quad \Longrightarrow \quad \int \frac{1}{P}dP = \int -\frac{g}{R_\alpha T}dz$$
$$\Longrightarrow \quad P(z) = P_0 \exp\left(-\frac{g}{R_\alpha T}z\right)$$

- Hydrostatic Equation, with varying temperature requires integration by substitution:

$$\frac{dP}{dz} = -\frac{P}{R_a\left(T_{surface} - 0.01z\right)}g \quad \Longrightarrow \quad \int \frac{1}{P}dP = -\frac{g}{R_a}\int \frac{1}{T_{surface} - 0.01z}dz$$
$$\Longrightarrow \quad P(z) = P_0 \exp\left(\frac{100g}{R_a}\ln\left(\frac{T_{surface} - 0.01z}{T_{surface}}\right)\right)$$

- **Second-order ordinary differential equations** are ODEs that involve a **second-order derivative**.
- We could devote an entire course looking at solving different types of second-order ODEs, but we will only concentrate on

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$$

where a, b and c are constants.

- This is referred to as a **homogeneous, linear, second-order ODE**.
- Note that we can write the exact same expression using alternative notation as per the following equation:

$$ay'' + by' + cy = 0$$

- To solve these second-order ODEs we first write out the **auxiliary** equation:

$$a\lambda^2 + b\lambda + c = 0$$

- Find roots of the quadratic equations using:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- If λ_1 and λ_2 are real and not equal ($\lambda_1 \neq \lambda_2$ then we can write a general solution as:

$$y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$$

- If our two roots are real and equal (i.e. $\lambda_1 = \lambda_2$) then we use the formula:

$$y = (A + Bx)e^{\lambda_1 x}$$

- To find the values of the constants A and B, we use information such as $y = 0$ and $y = 1$ when $x = 0$.
- These are called **initial conditions**, and they tell us the value of the solution and/or its derivatives at a specific point, allowing for a unique solution to the differential equation.
- This will lead us to solving simultaneous equations.
- Once we

- Fick's and Fourier's Laws describe the transfer of mass and heat.
- If we assume steady state, we get second-order ODEs which we can simply integrate twice

$$\frac{d^2T}{dx^2} = 0 \implies T(x) = Ax + B$$
$$\frac{d^2C}{dx^2} = 0 \implies C(x) = Cx + D$$

- We can use boundary conditions to solve this and find an expression.
- **Boundary value problems** differ from **initial value problems** as we specify upper and lower boundary conditions are specified (i.e, when $x = x_0$ and $x = x_1$).
- This gives us more information this means that we have to make the solution meet at both points.

2023-24 Past Paper

- Select the correct integral (i.e. $\int f(x)dx$) of the following function ($f(x)$) from the list given below. Please show your working:

$$4x^3 - 2x + 5$$

(a) $x^4 - x^2 + 5x + C$
(b) $x^4 - 2x^2 + 5x + C$
(c) $x^4 - x^2 + 5 + C$
(d) $3x^4 - x^2 + 5x$

- Perform indefinite integration:

$$\begin{aligned}\int 4x^3 - 2x + 5\ dx &= \int 4x^3 dx - \int 2x dx + \int 5 dx \\ &= \frac{1}{4} \times 4x^4 - \frac{1}{2} \times 2x^2 + 5x + C \\ &= x^4 - x^2 + 5x + C\end{aligned}$$

- So the answer is (a).

2023-24 Past Paper

- Integrate the following expression and give your answer to 3 decimal places:

$$\int_{-3}^{4} f^2 - e^{3f}\, df$$

- Perform definite integration:

$$\begin{aligned}
\int_{-3}^{4} f^2 - e^{3f}\, df &= \int_{-3}^{4} f^2\, df - \int_{-3}^{4} e^{3f}\, df \\
&= \frac{1}{3} \times \left[f^3\right]_{-3}^{4} - \frac{1}{3}\left[e^{3f}\right]_{-3}^{4} \\
&= \frac{1}{3} \times (4^3 - (-3)^3) - \frac{1}{3}(e^{3\times 4} - e^{3\times(-3)} \\
&= \frac{91}{3} - \frac{e^{12} - e^{-9}}{3} \\
&= -54,221.264
\end{aligned}$$

- A body of mass m kilograms rests on a slope that has an angle of inclination of 30 degrees.
- Calculate the speed of this body after 10 seconds, assuming the initial velocity is $0m/s$ and the earths gravitation constant is $g = 9.8$. For this problem you need to write all the forces acting on the body in the form of Newtons second law:

$$F = ma = m\frac{dv}{dt}$$

Where F is the total force acting on a body of mass m, leading to the change in velocity $\frac{dv}{dt}$. You can then integrate this equation and evaluate it the given initial conditions

- The total forces acting towards the earth is $F = mg$ but as it is in on an incline, we need to take account for this. Using trigonometry, we can write these as follows:

$$F = mg\sin(30)$$

- Write out the full ODE:

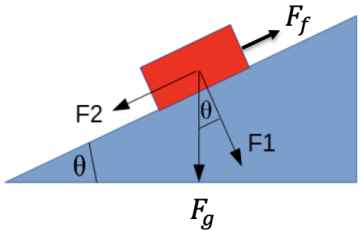
$$\begin{aligned} m\frac{dv}{dt} &= mg\sin(30) \\ \frac{dv}{dt} &= g\sin(30) \end{aligned}$$

- At time $t = 0$, velocity is $0m/s$ and at time 10, velocity is v. Use separation of variables and integrate:

$$\begin{aligned} dv &= g\sin(30)dt \\ \int_0^v dv &= g\sin(30)\int_0^{10} dt \\ [v]_0^v &= g\sin(30)\,[t]_0^{10}\,dt \\ v &= 10g\sin(30) = 49m/s \end{aligned}$$

2023-24 Past Paper

- At time $t = 10$, a force is suddenly applied to the body but opposite to the direction of travel.
- We are told this force is directly proportional to the mass m via a coefficient α where $\alpha = 8.5$.
- The new force is thus given as $F = \alpha m$. Draw a schematic diagram that displays all the forces now acting on the body
- Determine how long it takes for the vehicle to stop, to 2 decimal places.



$F_g = mg$
$\theta = 45^o$
$F1 = mg \cdot \cos(\theta) = mg \cdot \cos(45)$
$F2 = mg \cdot \sin(\theta) = mg \cdot \sin(45)$
$F_f = \alpha m$

- There is a force $F = \alpha m$ now acting in the opposite direction of travel so the total forces acting on the object are:

$$F = mg\sin(30) - \alpha m$$

- Write out the new ODE:

$$\begin{aligned} m\frac{dv}{dt} &= mg\sin(30) - \alpha m \\ \frac{dv}{dt} &= g\sin(30) - \alpha \end{aligned}$$

- At time $t = 0$, velocity is $49m/s$ and at time t, velocity is $0m/s$. Use separation of variables and integrate:

$$\begin{aligned} dv &= (g\sin(30) - \alpha)dt \\ \int_{49}^{0} dv &= (g\sin(30) - \alpha)\int_{0}^{t} dt \\ [v]_{49}^{0} &= (g\sin(30) - \alpha)\,[t]_{0}^{t}\,dt \\ -49 &= (g\sin(30) - \alpha)t \\ -49 &= -3.6t \\ t &= 13.61 \end{aligned}$$

- The following equation described the rate of change in a population P

$$\frac{dP}{dt} = \alpha P - \beta P$$

  Where α is a coefficient that determines a contribution according to births, whilst β a coefficient that determines the contribution from natural deaths.

- Integrate the equation to arrive at the following equation that describes how P varies with time, if P_0 represents the population value when $t = 0$:

$$P(t) = P_0 \exp((\alpha - \beta)t)$$

2023-24 Past Paper

- Use separation of variables and indefinite integration:

$$\begin{aligned} \frac{1}{P}dP &= (\alpha - \beta)dt \\ \int \frac{1}{P}dP &= (\alpha - \beta)\int dt \\ \ln(P) &= (\alpha - \beta)t + C \end{aligned}$$

- At time $t = 0$, $P = P_0$:

$$\ln(P_0) = C$$

- Substitute and rearrange:

$$\begin{aligned} \ln(P) &= (\alpha - \beta)t + \ln(P_0) \\ P(t) &= \exp((\alpha - \beta)t + \ln(P_0)) \\ &= P_0 \exp((\alpha - \beta)t) \end{aligned}$$

- We are now told the birth rate is constant, such that:

$$\frac{dP}{dt} = \psi - \beta P$$

Where ψ represents this constant. To integrate this expression, you will need to use integration using a substitution, creating a new variable $u = \psi - \beta P$ equal to arrive at the following expression:

$$P(t) = \frac{1}{\beta}\left(\psi - (\psi - \beta P_0)\exp(-\beta t)\right)$$

- Use separation of variables:

$$\begin{aligned} \frac{1}{\psi - \beta P} dP &= dt \\ \int \frac{1}{\psi - \beta P} dP &= \int dt \end{aligned}$$

- Let's use $u = \psi - \beta P$ as a substitution. Lets rearrange for P, differentiate in terms of u and find an expression for dP:

$$u = \psi - \beta P \implies P = \frac{1}{\beta}(u - \psi) \implies \frac{dP}{du} = -\frac{1}{\beta} \implies dx = -\frac{1}{\beta} du$$

- Use these to make the substitution:

$$\int \frac{1}{\psi - \beta P} dP \implies -\frac{1}{\beta} \int \frac{1}{u} du$$

- Now we can evaluate using the basic laws of integration:

$$\begin{aligned} \frac{1}{\beta}\int \frac{1}{u}du &= -\int dt \\ \frac{1}{\beta}\ln(u) &= -t + C \end{aligned}$$

- At time $t = 0$, $u = U_0 = \psi - \beta P_0$:

$$\frac{1}{\beta}\ln(U_0) = C$$

- Substitute and rearrange:

$$\begin{aligned} \frac{1}{\beta}\ln(u) &= -t + \frac{1}{\beta}\ln(U_0) \\ \ln(u) &= -\beta t + \ln(U_0) \\ \ln(u) - \ln(U_0) &= -\beta t \\ \ln\left(\frac{u}{U_0}\right) &= -\beta t \end{aligned}$$

- Substitute $u = \psi - \beta P$ and $U_0 = \psi - \beta P_0$ and rearrange further

$$\begin{aligned}
\ln\left(\frac{u}{U_0}\right) &= -\beta t \\
\frac{u}{U_0} &= \exp(-\beta t) \\
\frac{\psi - \beta P}{\psi - \beta P_0} &= \exp(-\beta t) \\
\psi - \beta P &= (\psi - \beta P_0)\exp(-\beta t) \\
\beta P &= \psi - (\psi - \beta P_0)\exp(-\beta t) \\
P &= \frac{1}{\beta}(\psi - (\psi - \beta P_0)\exp(-\beta t))
\end{aligned}$$

- The hydrostatic equation allows us to calculate the variation in pressure P with height z in the atmosphere:

$$\frac{dP}{dz} = -\frac{g}{R_\alpha(T_s - \alpha z)}P$$

  Where $R_\alpha = 287 Jkg^{-1}K^{-1}$. We have been told that the value for α could be either 0.01 or 0.02. We want to know what difference this makes in the final pressure at a height of 1400m.

- Integrate the hydrostatic equation using a substitution, when $\alpha = 0.01$. This should lead you to derive the following value for the pressure at 1400m:

$$P(z) = 100000 \exp\left(\frac{g}{0.01 \times 287} \ln\left(\frac{276}{290}\right)\right) \approx 84454 Pa$$

- At $z = 0$, $P = 100000$ and at $z = 1400$, $P = P$. Use separation of variables:

$$\int_{100000}^{P} \frac{1}{P} dP = -\frac{g}{R_\alpha} \int_0^{1400} \frac{dz}{(T_s - \alpha z)}$$

- Use integration by substitution with $u = T_s - \alpha z$ as a substitution. Lets rearrange for z, differentiate in terms of u and find an expression for du:

$$u = T_s - \alpha z \implies z = \frac{1}{\alpha}(u - T_s) \implies \frac{dz}{du} = -\frac{1}{\alpha} \implies dz = -\frac{1}{\alpha} du$$

- Alter the limits:

$$z = 0 \implies u = T_s - \alpha \times 0 = T_s$$
$$z = 1400 \implies u = T_s - 1400\alpha$$

- Use these to make the substitution:

$$\int_0^{1400} \frac{dz}{(T_s - \alpha z)} \implies -\frac{1}{\alpha} \int_{T_s}^{T_s - 1400\alpha} \frac{du}{u}$$

- Now we can evaluate using the basic laws of integration:

$$\begin{aligned}
\int_{100000}^{P} \frac{1}{P} dP &= \frac{g}{\alpha \times R_\alpha} \int_{T_s}^{T_s - 1400\alpha} \frac{du}{u} \\
[\ln(P)]_{100000}^{P} &= \frac{g}{\alpha \times R_\alpha} [\ln(u)]_{T_s}^{T_s - 1400\alpha} \\
\ln(P) - \ln(100000) &= \frac{g}{\alpha \times R_\alpha} (\ln(T_s - 1400\alpha) - \ln(T_s))
\end{aligned}$$

- Rearrange for final expression:

$$\begin{aligned}
\ln(P) - \ln(100000) &= \frac{g}{\alpha \times R_\alpha}(\ln(T_s - 1400\alpha) - \ln(T_s)) \\
\ln\left(\frac{P}{100000}\right) &= \frac{g}{\alpha \times R_\alpha} \ln\left(\frac{T_s - 1400\alpha}{T_s}\right) \\
\frac{P}{100000} &= \exp\left(\frac{g}{\alpha \times R_\alpha} \ln\left(\frac{T_s - 1400\alpha}{T_s}\right)\right) \\
P &= 100000 \exp\left(\frac{g}{\alpha \times R_\alpha} \ln\left(\frac{T_s - 1400\alpha}{T_s}\right)\right)
\end{aligned}$$

- Plugging in $\alpha = 0.01$, $T_s = 290$ and $R_\alpha = 287$ gives us:

$$P = 100000 \exp\left(\frac{g}{0.01 \times 287} \ln\left(\frac{276}{290}\right)\right)$$

- What is the percentage difference in pressure at 1400m when we change $alpha = 0.01$ to $\alpha = 0.02$. Use 5 significant digits for the value of P at 1400m as we have done for part (a) and 1 decimal place for the percentage difference.

- In the previous part we arrived at the solution:

$$P = 100000 \exp\left(\frac{g}{\alpha \times R_\alpha} \ln\left(\frac{T_s - 1400\alpha}{T_s}\right)\right)$$

- Plugging in $\alpha = 0.02$, $T_s = 290$ and $R_\alpha = 287$ gives us:

$$P = 100000 \exp\left(\frac{g}{0.02 \times 287} \ln\left(\frac{262}{290}\right)\right) = 84083Pa$$

- Percentage difference is given by:

$$100 \times \frac{84083 - 84454}{84454} = -0.4\%$$